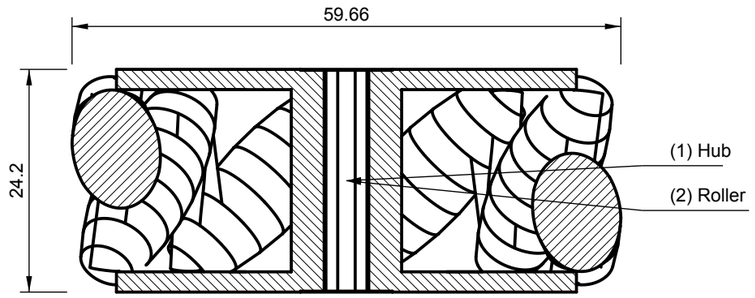
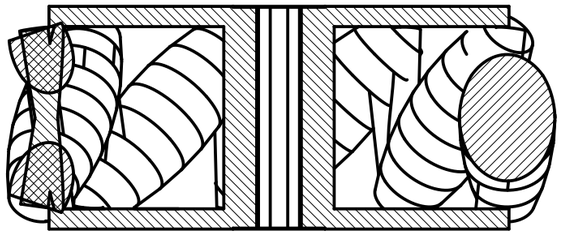
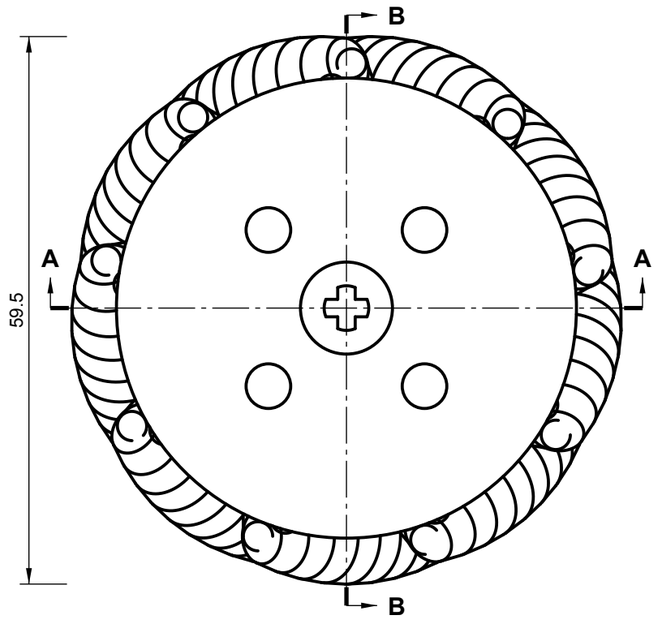
A–A



B–B

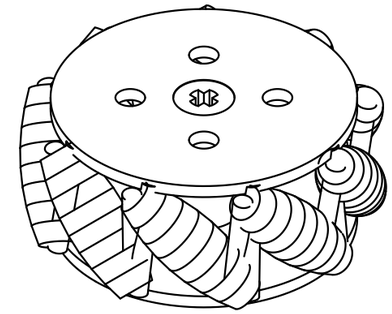




Bill of materials 明细表

#	Part No. 零件号	Name 名称	Material 材料	Colour	Qty	g
1	SB-WM60R-01	Hub 轮毂	ABS	Blue	1	13.45
2	SB-WM60R-02	Roller 滚轮	TPU	Black	9	17.80

ISOMETRIC 轴测图 (1:1, ref.)



Technical requirements 技术要求

1. Assembly drawing: components per BOM, each with its own drawing (sheets 2…).
装配图：零件见明细表，各零件另有图纸（第 2 页起）。

2. Tyre / rollers fitted on the hub; wheel must run true within 0.3 TIR.
轮胎/滚轮装于轮毂上，径向跳动≤0.3。

3. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

4. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

5. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
第二阶段图纸用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	ABS + TPU see BOM 见明细表	**Weight** 重量	31.3 g
Colour 颜色	see BOM 见明细表	**Qty / kit** 每套数量	2
Finish 表面	see BOM 见明细表	**Scale** 比例	1.5:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	1 / 3

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

Rev 版本	Description 修改内容	Date 日期
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Part No. 零件号

SB-WM60R

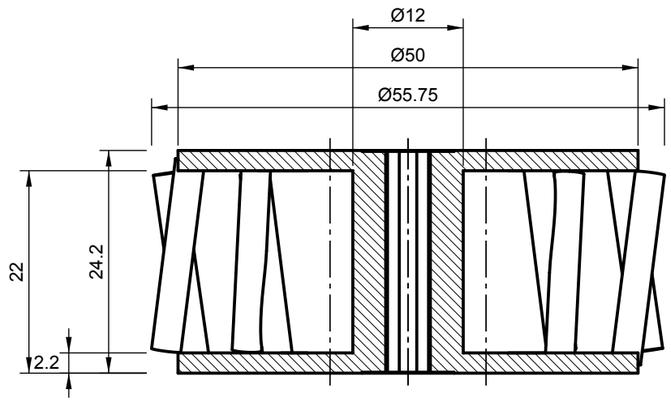
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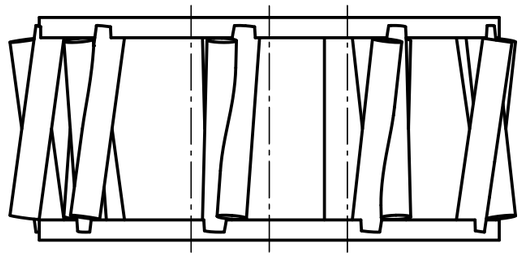
Mecanum wheel 60 (right) — assembly 装配图

麦克纳姆轮 60（右）

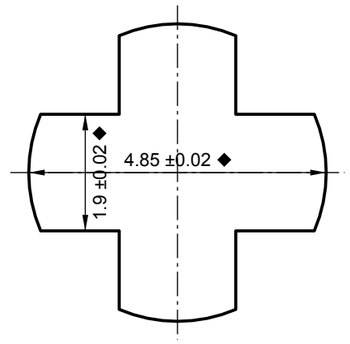
Dwg SB-WM60R · Rev A · A3 · 3D: SB-WM60R.step / .FCStd

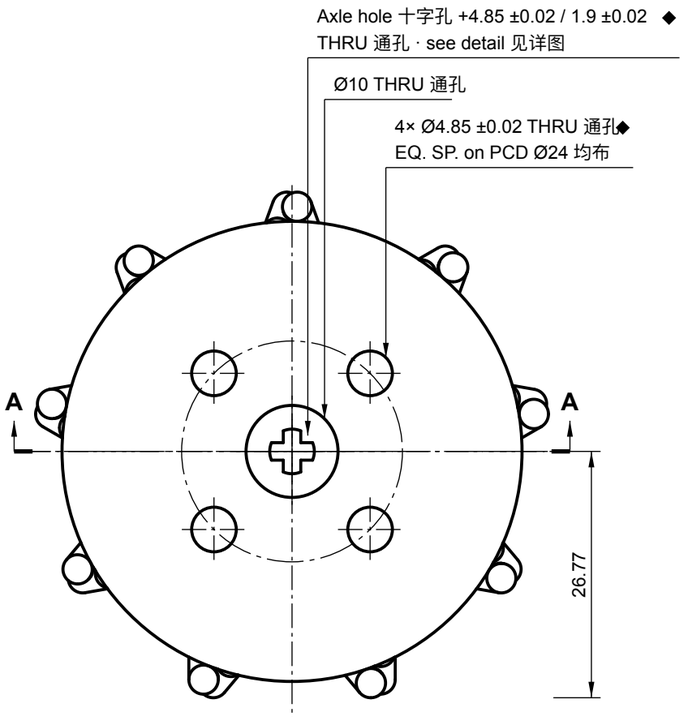
A–A



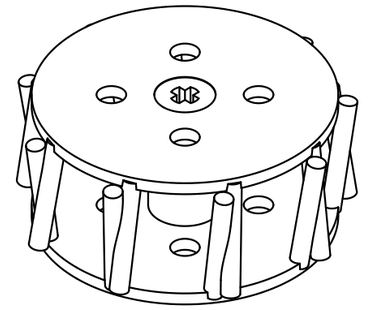


Axle hole 十字孔 (10:1)





ISOMETRIC 轴测图 (1:1, ref.)



Technical requirements 技术要求

1. 3D model SB-WM60R.step is the master; dimensions not shown are taken from it.
以 3D 模型 SB-WM60R.step 为准，未注尺寸按 3D 数据。

2. Material ABS, e.g. CHIMEI PA-757 or equiv.; colour Blue 蓝 (sRGB #1F6FE5, std SB-C01); colour plaque approved before production.
材料：ABS（奇美 PA-757 或同等）；颜色 蓝（#1F6FE5，色号 SB-C01），量产前确认色板。

3. Draft ≤ 0.5° on outer walls, added by the tool maker within tolerance. Pin / axle holes and ◆ features: draft ≤ 0.25°, ◆ size held over the full depth.
外壁拔模≤0.5°（模具厂在公差内添加）；销孔、十字孔及◆特征拔模≤0.25°，全深度保持◆尺寸。

4. Parting line, gate and ejector marks: not on visible faces or inside holes, flush to +0.05. Positions to be agreed in DFM.
分模线、浇口、顶针痕不得在外观面及孔内，凸起≤0.05，位置于 DFM 时确认。

5. No flash > 0.05, sink marks, short shots, silver streaks or voids. Holes free of burrs.
无大于0.05的披锋，无缩水、缺料、银纹、气泡；孔内无毛刺。

6. Unspecified edge radii R0.2–R0.35 as in 3D. No sharp edges or points (toy safety).
未注圆角 R0.2–R0.35（按 3D），不得有锐边锐角（玩具安全）。

7. Pin holes: GO Ø4.83 / NO-GO Ø4.88 plug gauge. Fit test with SemiBlock and LEGO® Technic™ pins; push-in / pull-out force agreed at T1.
销孔用 Ø4.83 通规 / Ø4.88 止规检验；与 SemiBlock 及乐高销做装配测试，插拔力于 T1 确认。

8. Axle (+) holes: GO gauge = cross 4.83 / 1.88. Standard axle must slide in by hand without play > 0.05.
十字孔用 4.83/1.88 十字通规；标准十字轴可手推插入，间隙≤0.05。

9. Mark cavity number and material code >ABS< on a hidden face where space allows (ISO 11469).
在不可见面标注穴号及材料代码 >ABS<（ISO 11469），空间不足可省略。

10. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

11. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

12. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.

Material 材料	ABS (PA-757 or equiv.) ABS 奇美 PA-757 或同等	**Weight** 重量	13.5 g
Colour 颜色	Blue 蓝 #1F6FE5	**Qty / kit** 每套数量	2
Finish 表面	VDI 3400 Ref 18 细哑光纹理	**Scale** 比例	1.5:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	2 / 3

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

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Approved 批准		

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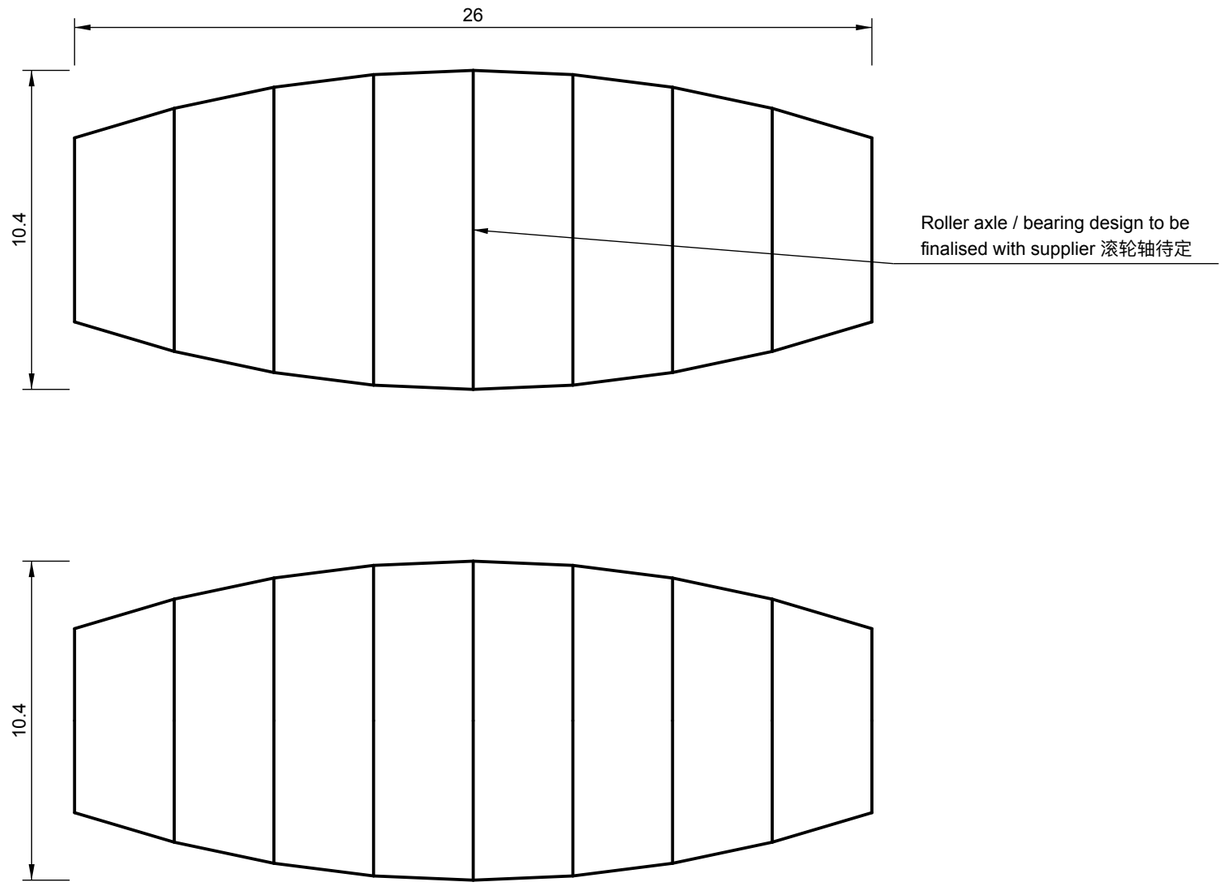
Part No. 零件号

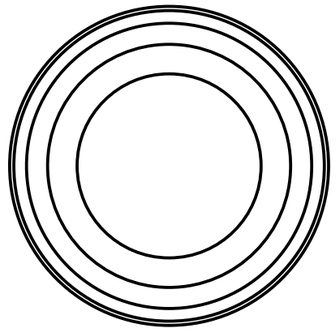
SB-WM60R-01

Name 名称

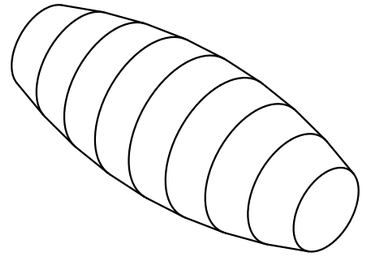
Mecanum wheel 60 (right) — Hub
麦克纳姆轮 60（右） — 轮毂

Dwg SB-WM60R-01 · Rev A · A3 · 3D: SB-WM60R.step / .FCStd





ISOMETRIC 轴测图 (2.5:1, ref.)



Technical requirements 技术要求

1. 3D model SB-WM60R.step is the master; dimensions not shown are taken from it.
以 3D 模型 SB-WM60R.step 为准，未注尺寸按 3D 数据。

2. Material TPU, Shore A 85 ±5; colour Black 黑 (sRGB #1C1F23, std SB-C10); colour p laque approved before production.
材料：TPU 邵氏 A 85±5；颜色 黑（#1C1F23，色号 SB-C10），量产前确认色板。

3. Draft ≤ 0.5° on outer walls, added by the tool maker within tolerance. Pin / axle holes and ◆ features: draft ≤ 0.25°, ◆ size held over the full depth.
外壁拔模≤0.5°（模具厂在公差内添加）；销孔、十字孔及◆特征拔模≤0.25°，全深度保持◆尺寸。

4. Parting line, gate and ejector marks: not on visible faces or inside holes, flush to +0.05. Positions to be agreed in DFM.
分模线、浇口、顶针痕不得在外观面及孔内，凸起≤0.05，位置于 DFM 时确认。

5. No flash > 0.05, sink marks, short shots, silver streaks or voids. Holes free of burrs.
无大于0.05的披锋，无缩水、缺料、银纹、气泡；孔内无毛刺。

6. Unspecified edge radii R0.2–R0.35 as in 3D. No sharp edges or points (toy safety).
未注圆角 R0.2–R0.35（按 3D），不得有锐边锐角（玩具安全）。

7. Mark cavity number and material code >TPU< on a hidden face where space allows (ISO 11469).
在不可见面标注穴号及材料代码 >TPU<（ISO 11469），空间不足可省略。

8. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

9. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

10. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
第二阶段图纸用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	TPU Shore A 85 TPU 邵氏 A85	**Weight** 重量	1.98 g
Colour 颜色	Black 黑 #1C1F23	**Qty / kit** 每套数量	18
Finish 表面	VDI 3400 Ref 24 哑光纹理	**Scale** 比例	5:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	3 / 3

General tolerances 未注公差 (mm) rigid ABS/POM 硬胶 | TPE 软胶		
0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

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Part No. 零件号

SB-WM60R-02

Name 名称

Mecanum wheel 60 (right) — Roller (×9)

麦克纳姆轮 60（右） — 滚轮

Dwg SB-WM60R-02 · Rev A · A3 · 3D: SB-WM60R.step / .FCStd